2024 AI Technology Report

This report provides an overview of major AI developments in 2024. Large Language Models (LLMs) have seen significant improvements in reasoning capabilities, multilingual understanding, and tool use.

Key Metrics Comparison

Model	Parameters	Context Length	MMLU Score
GPT-4o	1.8T (est.)	128K	88.7%
Claude 3.5	Unknown	200K	88.3%
Gemini Ultra	Unknown	1M	90.0%
Llama 3.1	405B	128K	85.2%

The table above shows that Gemini Ultra achieves the highest MMLU score at 90.0%, while Claude 3.5 offers the longest context window at 200K tokens. Open-source models like Llama 3.1 are closing the gap with proprietary models.

Retrieval-Augmented Generation

RAG (Retrieval-Augmented Generation) combines information retrieval with text generation. The system first retrieves relevant documents from a knowledge base, then uses an LLM to generate answers grounded in those documents.

Key RAG improvements in 2024:
- Hybrid search combining BM25 and vector retrieval
- Cross-encoder reranking for higher precision
- Multi-modal RAG handling tables, images, and structured data
- Query rewriting and expansion techniques